

JavaScript Validation

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JavaScript Form Validation

HTML form validation can be done by JavaScript.

If a form field (fname) is empty, the `required` attribute prevents form submission:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Form Validation</h4>
<form action="/action_page.php" method="post">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname" required>
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

Example 1

Result [View Example](#)

JavaScript Can Validate Numeric Input

JavaScript is often used to validate numeric input:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Validate Numeric Input</h4>
<p>Please input a number between 1 and 10:</p>
<input id="numb">
<button type="button" onclick="myFunction()">Submit</button>
<p id="demo"></p>

<script>
function myFunction() {
  let x = document.getElementById("numb").value;
  let text;
  if (isNaN(x) || x < 1 || x > 10) {
    text = "Input not valid";
  } else {
    text = "Input OK";
  }
  document.getElementById("demo").innerHTML = text;
}
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Automatic HTML Form Validation

HTML form validation can be performed automatically by the browser using the `required` attribute:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Validation</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Automatic HTML Form Validation</h4>
<form action="/action_page.php" method="post">
  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>
  <input type="submit" value="Submit">
</form>
<p>The required attribute prevents submission if the field is empty.</p>

</body>
</html>
```

Example 3

Result [View Example](#)

Data Validation

Data validation is the process of ensuring that user input is clean, correct, and useful.

Typical validation tasks:

- Has the user filled in all required fields?
- Has the user entered a valid date?
- Has the user entered text in a numeric field?
- Has the user entered a valid email address?

Constraint Validation HTML Input Attributes

Attribute	Description
<code>required</code>	Specifies that an input field must be filled in
<code>min</code> / <code>max</code>	Specifies min/max value of an input field
<code>minlength</code> / <code>maxlength</code>	Specifies min/max length of text input
<code>type</code>	Specifies the type of input (email, number, etc.)
<code>pattern</code>	Specifies a regex pattern for input validation

HTML Constraint Validation

Constraint validation in HTML is based on:

- **Constraint validation HTML Input Attributes** - described above
- **Constraint validation CSS Pseudo Selectors** - style form elements based on validity:
 - `:valid` - Selects form elements with a valid value
 - `:invalid` - Selects form elements with an invalid value

Example: Input fields with `type="email"` will automatically be validated when the form is submitted. The `:valid` and `:invalid` CSS pseudo-classes allow you to style valid/invalid fields.

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Validation](#)